Chapter 43
An Artificial Intelligence and Computer Vision Based EyeWriter



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Abstract The authors in this paper devise a low-cost EyeWriter that can be used as an assisting aid for patients. The proposed system aims to give voice to the voiceless persons who have been living as an isolated section of the society due to non-existence of any efficient system. In the proposed system, the user communicates by writing the alphabet by alphabets in order to form words and sentences through his/her eyes. The proposed system uses artificial intelligence, OpenCV, Python, Python frameworks, dlib, neural networks, and computer vision. The suggested system is also cost-effective as it does not require any specialized hardware. The proposed model is also simulated and the obtained results are quite encouraging and hence the proposed system can be considered for its widespread deployment in the real world.

43.1 Introduction

Many times, we come across many people who are unable to talk and express themselves owing to some disease or accident [1]. As a result, these patients experience

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the ignorance of the society toward them which further worsens their health condition. Considering the huge section of society which is experiencing any such disease preventing them from communicating, it becomes an issue of paramount importance to design an effective system that gives them voice. According to authors in [2], it is estimated that a sound 5.4 million persons are living with full or partial paralysis. Among these, around 72.1% of people are younger than 65 years. Hence, it is realized that an efficient and effective communication system must be designed so as to give voice to this ignorant and inarticulate segment of the society. Additionally, such systems can also prove to be quite effective during this pandemic time when the whole world is trying to control the spread through social distancing [3, 4].

In order to give voice to these otherwise dumb people, there have been several attempts using eye movement. The current system is also based on the fact that such people still have the ability to voluntarily control their eyes, and hence detects the gaze of the patient that enables him to communicate with the surrounding world. Thus, the tracking system acts as an assistance to the disabled persons. During past few years, several researchers have been working to provide similar kinds of assistance to physically disabled persons. The prime focus of such research groups is to provide such novel assistive technology at an affordable cost. From the literature, it becomes evident that there are several commercial products available in the market. However, the majority of such products are beyond the reach of a huge segment of disabled persons owing to various factors.

Furthermore, there remains a huge section of concerned society that can be catered by providing an efficient and cost-effective solution. According to authors in [5, 6], patients of amyotrophic lateral sclerosis (ALS) and paralysis, i.e., people having severe physical disabilities, have been using assistive tools like EyeWriter. Unfortunately, the state-of-the-art eye tracking system is beyond the affordability of most of the people, and hence, they have been experiencing the need of an alternative system. This escalated cost of the present system is due to the limited sale of the eye tracking system during the year. Hence, it becomes imperative to design an economical solution in order to ensure its widespread usage. For the same, it necessitates to have dedicated efforts. The authors in this chapter attempt to provide such a solution in a cost effective manner.

The paper has been organized into various sections. The need of an efficient EyeWriter system is discussed in the introductory Sect. 43.1. Related work has been presented in Sect. 43.2. Section 43.3 elaborates the proposed work and the implementation and results have been demonstrated in Sect. 43.4. Further, concluding comments and future work are discussed in Sect. 43.5.

43.2 Related Work

As mentioned earlier that number of research communicated have been lured to work in the field of providing assistance to physically disabled persons as a result of technological revolutions in the related field. For instance, the revolutionary transformations

in the domain of AI, computer vision, and ML have resulted in the evolution in the medical field particularly in the direction of devising efficient assistance devices [7, 8]. Here, authors discuss the methods suggested by different researchers to detect eye tracking and eye gazing. International teams are also working to devise efficient methods that enable ALS patients to draw with their eyes.

The first eye tracking device was bit invasive and uncomfortable. This was resolved by authors in the year 1901. Authors in [9] have designed the first modest model of eye movements that uses light reflections from the corners of corners and a photographic method [10]. However, there are several other technologies in existence to capture eye movements. In such methods, the invasiveness and accuracy depends on the method and device employed. These methods are broadly classified into four classes, namely, scleral search coil method (SSC), video-occulography (VOG), infrared-occulography (IROG), and electro-occulography (EOG). Authors in this paper refrain from elaboration of such approaches. However, interested readers may refer to work by authors in [11, 12] for detailed understanding of given classifications.

Generally, VOG method fetches image of eye using a camera which is generally mounted at some remote location or head gear. The captured image is used to extract information through various eye features to evaluate the point of gaze (POG) [13]. This method has gained widespread popularity as the VOG is capable to minimize the invasiveness to a large extent. Interested readers may also refer to www.eyewriter.org where the members of Free Art and Technology (FAT), and The Ebeling Group communities worked together to devise a tool named TEMPTONE. The general model of the eye tracking system tracks the position of eye pupil through camera and uses the same to position the cursor on a computer screen and/or projector [11].

From the related literature, it is evident that eye tracking technique has gained widespread attention and has become a prime technique for human and computer interaction. Hence, continuous work is taking place in the direction of devising efficient assistant tools and methods for people having speech and movement disabilities. One motivating and exciting part is that such eye tracing devices are gradually getting cheaper as a result of industrial and technological revolution, bringing it in the reach of maximum people. Authors in this paper also aim to propose a cost-effective solution for eye tracking as described in the following section.

43.3 Proposed Methodology

The proposed method is dedicated to design a relatively inexpensive eye tracking system. In the proposed system, the traditional webcam of the laptop can be used, and hence, it does not require any extra hardware. The proposed approach is competent to correctly perform classification (for positioning the pupil of eye) even in the presence of noise. In the proposed approach, the cursor is shown along different keys on the virtual keyboard. This cursor keeps on moving throughout the keyboard, and by detecting the eye blink, the key of interest is detected. Thus, the proposed system allows users to type a single character at a time which can be combined to form



Fig. 43.1. 68 Face landmarks set

words and sentences. The selected characters are demonstrated on the white board which is also shown to the user on the screen.

The proposed methodology has been divided into two phases. Here, phase 1 basically focus on the detection of eye, blinking of eye, and its gaze, whereas phase 2 handles virtual keyboard, pressing of a key through eye blink, playing sounds, and representation of input through white dialog box.

During phase 1, the proposed model makes use of state-of-the-art landmarks models to detect 68 specific landmarks of the face as shown in Fig. 43.1. A specific index is assigned to each point in these landmarks. From Fig. 43.1, it is evident that left eye points are from 36 to 41 whereas right eye is pointed by points 42 to point 47.

Detection of eye blink uses horizontal and vertical lines as shown in Fig. 43.2. Now, it is evident from Fig. 43.2 that if the eye is open, the length of horizontal line nearly matches with that of vertical line. On the contrary, the length of vertical line is much smaller than that of horizontal line when the eye is closed. Thus, eye blink model detects whether eye is closed or not considering these observation. In the proposed method, horizontal line acts as the point of reference, and then the ratio of vertical line to this reference line is calculated. If the ratio goes below a predetermined threshold value, the model will consider the eye to be closed, otherwise open.

The next part of the proposed model is to detect the eye Gaze. For the same, a virtual keyboard is illustrated on the computer screen. The keyboard shown on the screen is segmented into two halves, i.e., the left side and right side as shown in Fig. 43.3. In the shown keyboard, each key in turn is lighted up for few seconds.

Phase 2 works by considering the eye split into two parts viz. left part and right part. It finds out in which of the two parts of the keyboard, there is more sclera (white of the eye) visible. Technically, to detect the sclera, eye is converted into grayscale and the white pixels are determined. Thereafter, determined white pixels are compared with a threshold value to obtain gaze ratio. The gaze ratio indicates where the specific eye is looking. Normally, both the eyes look in the same direction, so if gaze of a single eye is detected correctly, the gaze of both eyes can also be determined efficiently and correctly as shown in Fig. 43.4.

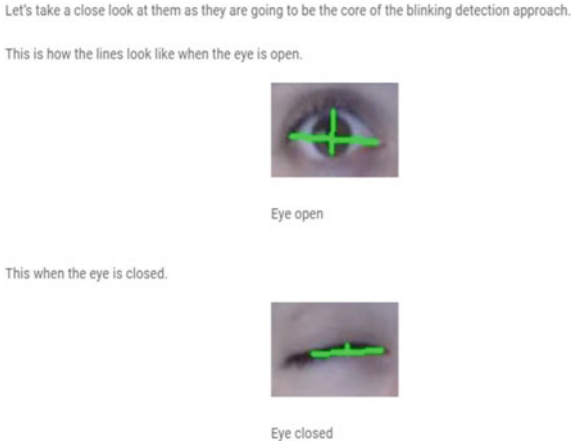


Fig. 43.2 Horizontal and vertical line representation



Fig. 43.3 Virtual keyboard segmented into two halves



(a) Sclera representation for various gaze (b) Gaze of single eye

Fig. 43.4 Illustration of sclera and gaze

43.4 Implementations and Results

In order to simulate the proposed cost effective eye-write, authors created a black screen with width of 1500 pixels and height of 1000 pixels. This black screen is created in order to simulate a virtual keyboard as shown in Fig. 43.5. Here, Fig. 43.5 represents the left half of the virtual keyboard. Along with generating the black screen, the proposed model also uses a white board in order to demonstrate the message that the user is typing through his eye gaze. The corresponding white board is also demonstrated in Fig. 43.6.

In the simulated mode, each character in the keyboard blinks for few seconds. When the required character blinks, the user blinks his eyes and the corresponding character is shown in the whiteboard as illustrated in Fig. 43.7.

The simulated model is observed to be an efficient method particularly considering its cost as it barely has any additional cost. Considering the efficiency and cost effectiveness, the model can be considered for widespread application in real life.

The proposed cost-effective EyeWriter model demonstrates that the eye tracker is able to determine eye gaze in permissible time. The proposed system has yielded an accuracy rate of error pixels at 21 points or 0.76 in degree of visual angle from calibration process. Further, EyeWriter has potential advantages such as reduced screen footprint and minimal distance between input and output areas. The proposed model can also be extended to implement eye typing.

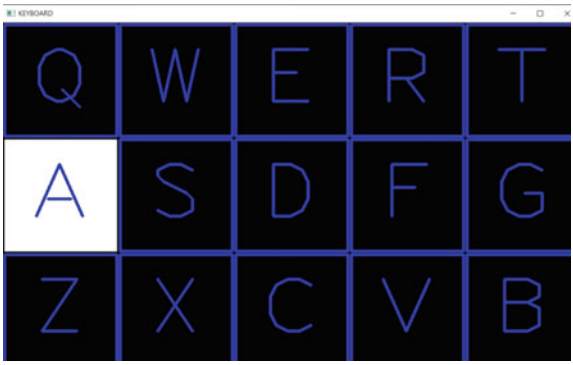


Fig. 43.5 Virtual keyboard (left half)



Fig. 43.6 White board for output representation

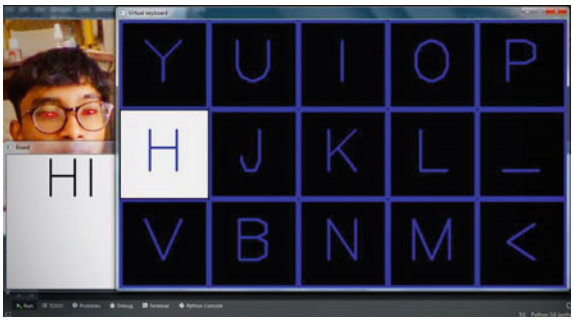


Fig. 43.7 Result through live video-stream of eye gaze detection in virtual keyboard

43.5 Conclusion and Future Work

EyeWriters have been considered to a luring and exciting area for research during the past few decades. This interest is escalated further due to technological revolutions in the allied fields such as artificial intelligence, computer vision, and Internet of Things. An efficient EyeWriter can prove to be a boon to the physically handicapped patients who can control the blink and movement of their eyes. Various authors have developed different models in this direction. Authors in this paper propose a cost effective solution for the same. The proposed model uses dlib in association with computer vision and outperforms the state-of-the-art models. The proposed system is evaluated in order to validate its effectiveness, and the results obtained are quite encouraging. This is further advocated considering the minimal additional cost of the proposed model. So, it can be concluded that the result obtained are motivating, and hence, the current work can be deployed in real life. The work can be further extended in the direction of guessing the complete word once the user types few initial characters of the word. Such a model will take the eye write to another level by further comforting its users.

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